

Hiep Tan To

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B.S. & M.S. Computer Science student at Ho Chi Minh University of Science, specializing in Machine Learning Engineering, Backend Development, and System Integration. Strong fundamentals in OOP, algorithms, and database systems. Experienced in designing and optimizing large-scale data/ML pipelines, deploying model-driven services, and contributing to open-source software.

Education

Master of Computer Science	In Progress
University of Science - VNUHCM	2025
Bachelor of Technology with Honors	9.41/10.0
Computer Science and Engineering, University of Science - VNUHCM	2021-2025
Thesis	10.0/10.0
Generative AI-Based Virtual KOL System	2025
Language Skills	IELTS 6.0, VSTEP B2

Internships and Experience

Junior AI Engineer

First AI

Aug. 2025 - May. 2026

- Developed scalable, real-time recommendation systems for AI-driven search and social platforms, significantly improving content relevance and user engagement.
- Trained and fine-tuned machine learning and LLM-based models on large-scale user behavior and contextual datasets to improve ranking accuracy.
- Implemented retrieval-augmented generation (RAG) pipelines to enhance recommendation accuracy with domain-specific external knowledge bases.
- Integrated external LLM APIs to support semantic search and personalized recommendations.
- Optimized model inference for real-time recommendation services in production with minimal latency.
- Evaluated recommendation and model performance using metrics such as Precision@K, Recall@K, NDCG, and F1-score.

Contractor Software Engineering

Integration Team, Treasure Data

Apr 2024 – Aug 2025

- Built and optimized data pipelines and connectors for ETL workflows (Embulk, REST APIs).
- Developed an AI document summarization agent, improving summarization speed by 30%.
- Collaborated with cross-functional teams to design scalable and reliable data integration.
- Enhanced CLI manual testing tools to support QA processes effectively.

Undergraduate Thesis: Generative AI-Based Virtual KOL System

Guide: Dr. Le Trung Nghia, Assoc. Prof. Tran Minh Triet, University Of Science

Sept. 2025

- Developed a generative AI system merging identity with varied styles, preserving semantics.
- Built a modular, extensible architecture for easy integration of new algorithms and pipelines.

Undergraduate Research

Software Engineering Laboratory, University of Science

Sept. 2023 – Present

- Researched Generative AI techniques such as Stable Diffusion, GANs, and RAG.
- Contributed to successful publications, enhancing understanding of adversarial robustness and model defense strategies for NLP systems.
- Conducted research on physics-aware video generation, focusing on modeling physical phenomena such as melting and spring-mass systems.
- Benchmarked and optimized OCR models, including Surya and olmOCR, to improve accuracy for Vietnamese handwriting recognition tasks.

Technical Skills

- **Machine Learning / Deep Learning:** PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, OpenCV
- **Generative AI:** Stable Diffusion, GANs, Retrieval-Augmented Generation (RAG), LangChain
- **Data Engineering / MLOps:** Embulk, Airflow, SQL, MongoDB, AWS (S3, EC2, Lambda)
- **Programming:** Python, C++, Java
- **Other Tools:** Streamlit, Docker, Git, Overleaf
- **Soft Skills:** Strong presentation, collaboration, and problem-solving

Scholastic Achievements

- **Graduated in the Top 5** of the cohort with **High Distinction** (GPA: 9.41/10.0, Perfect Thesis Score), earning **direct admission** to the Master's program in Computer Science at the University of Science.
- Awarded the prestigious **Certificate of Merit by the President of Vietnam National University** for exceptional academic and scientific research excellence.
- Awarded **Outstanding Student in Scientific Research Achievement** (2023).
- Honored in the **Top 100 Outstanding Students of Vietnam National University** (2023).
- Ranked **1st Best Student of Regular Program, Faculty of IT** (2021).
- Selected among **Top 5 Students** of the Faculty of IT (2021).
- Won **Third Prize** – Provincial Competition for Excellent Students in Informatics (2020).

Key Academic Projects

Machine-Generated Code Detection

Competitor | Kaggle & SemEval

2024

- Engineered a robust tabular dataset from raw source code by extracting Abstract Syntax Tree (AST) properties, Halstead complexity metrics, and syntactic structures.
- Engineered layout-based features and applied targeted data perturbations, such as systematically injecting whitespaces and newlines, to enhance the robustness of GPT detection models against formatting obfuscation.
- Developed an ensemble learning pipeline utilizing tree-based models (**XGBoost, LightGBM, CatBoost**) heavily optimized for high-dimensional structured tabular data.
- Implemented advanced feature selection and stratified cross-validation strategies to mitigate model overfitting and address severe class imbalances in the training distribution.
- Secured **2nd place** globally on the competition leaderboard, achieving a highly competitive **F1-score of 0.82**.

Question Answering System with Transformer Models

Researcher | Course Project

2024

- Developed a robust QA system by fine-tuning **BERT** and **RoBERTa** on **SQuAD, Zalo Challenge**, and custom Vietnamese datasets to enhance cross-lingual performance.
- Curated and pre-processed a large-scale dataset of **26k+ QA samples**; performed Exploratory Data Analysis (EDA) to mitigate class imbalance and improve data quality.
- Integrated **Google Custom Search API** to augment external knowledge, achieving a **25% reduction** in query response latency.
- Researched **Retrieval-Augmented Generation (RAG)** techniques by synthesizing insights from 100+ research papers to solve knowledge-intensive QA challenges.
- GitHub: [Question-Answering](#)

Adversarial Robustness in NLP Models

Researcher | Course Project

2024

- Evaluated the vulnerability of text classification models against adversarial attacks, including **FGSM, PGD, C&W**, and **TextFooler** using the **TextAttack** framework.
- Implemented an advanced **Adversarial Training** strategy, successfully recovering model accuracy from **0% to 95.24%** when under targeted adversarial attacks.
- Conducted extensive benchmarking on the **AG News** dataset, establishing a strong baseline with **75.12% accuracy** and **74.58 F1-score**.
- Analyzed the trade-offs between model performance and robustness to provide insights into securing production-level NLP deployments.
- GitHub: [Adversarial-Attacks](#)

Publication

GenKOL: Modular Generative AI Framework For Scalable Virtual KOL Generation

Team Size: 2

Jan. 2026

- Architected a scalable Generative AI framework built upon **Stable Diffusion** and custom-trained **LoRA** weights to automate the generation of diverse, high-fidelity Virtual KOLs fully.
- Engineered an ID-consistency preservation module utilizing advanced attention-injection and face-embedding techniques, ensuring stable visual identities across complex digital environments and dynamic poses.
- Optimized the end-to-end inference pipeline for large-scale deployment, significantly reducing character generation latency while maintaining photorealistic rendering quality.
- Peer-reviewed and accepted at **FJCAI 2026 - Can Tho** (The 3rd Future of Journalism and AI Conference).

Streamlining Virtual KOL Generation Through Modular Generative AI Architecture

Team Size: 2

Aug. 2025

- Designed a novel, multi-stage generative architecture integrating **ControlNet** and **IP-Adapter** to strictly control spatial structures and streamline the synthesis of highly realistic virtual human avatars.
- Implemented a robust, plug-and-play modular system supporting dense spatial conditioning (e.g., skeletal pose, depth, facial landmarks) to seamlessly switch between distinct visual styles and precise motion templates.
- Conducted comprehensive ablation studies demonstrating the superiority of the proposed modular architecture in achieving spatial accuracy and stylistic alignment over existing baselines.
- Presented research findings at **ACM Multimedia 2025** (Rank A*), showcasing significant computational efficiency advancements for AI-driven creative industries.

Budget-aware Road Semantic Segmentation in Unseen Foggy Scenes

Team Size: 3

Aug. 2023

- Proposed a highly efficient, budget-aware semantic segmentation model tailored for low-resource edge computing environments in autonomous driving systems.
- Leveraged Unsupervised **Domain Adaptation** alongside a Teacher-Student **Knowledge Distillation** framework to dynamically enhance model robustness against entirely unseen, heavily foggy weather conditions.
- Conducted extensive experiments on standard benchmark datasets (e.g., Foggy Cityscapes), achieving highly competitive Mean Intersection over Union (**mIoU**) scores without requiring computationally expensive retraining from scratch.
- Published and presented at the **RIVF 2023 International Conference**, demonstrating the model's capability to maintain real-time inference speed under severe weather degradation.
- Paper link: [IEEE Xplore](#)

Additional Certificates

- Completed **Generative AI for Educators with Gemini Certificate**
- Completed **FPT AI Compass**
- Completed **AWS Academy Cloud Foundations**
- Completed **Intro to Machine Learning (Kaggle)**
- Completed **Unsupervised Learning, Recommenders, Reinforcement Learning (Coursera)**
- Completed **Advanced Learning Algorithms (Coursera)**
- Completed **Machine Learning Explainability (Kaggle)**
- Completed **Machine Learning Specialization (Coursera)**
- Completed **Ultimate AWS Certified AI Practitioner AIF-C01**
- Completed **AWS Certified Developer Associate**